

Q1 Find

10 marks

Find the resource-management bug. `class Buffer { int* data; public: Buffer(int size) { data = new int[size]; } ~Buffer() { delete data; } };`

Your Answer

The bug is using `delete data` instead of `delete[]`. Since memory is allocated using `new[]`, we must use `delete[]`.

 Save Answer

Next →

Q2 Output

10 marks

What is the output? `class A { public: A() { cout << "A "; } ~A() { cout << "~A "; }}; int main() { A a1; { A a2; A a3; } A a4; }`

Your Answer

The output is AAAA~A~A~A~A because constructors execute in creation order and destructors execute in reverse order

 Save Answer

Next →

Q3 Problem

10 marks

What is the problem here? `class A { public: virtual void process() = 0; }; int main() { A obj; }`

Your Answer

Class A is an abstract class because it contains a pure virtual function.so we cannot create an object A

 Save Answer

Next →

Q4 Output

10 marks

What is the output? `class A { public: A() { cout << "A "; } ~A() { cout << "~A "; } }; class B : public A { public: B() { cout << "B "; } ~B() { cout << "~B "; } }; int main() { B obj; }`

Your Answer

The output is AB~B~A. The derived class constructor executes after the base constructor and destruction happens in reverse order.

 Save Answer

Next →

Q5 Identify

10 marks

Identify whether this is copy construction or assignment. A a; A b = a; A c; c = a;

Your Answer

A a; is Default construction
A b=a; is copy construction
A c; is Default construction
c=a is assignment

 Save Answer

Next →

Q6 Destructor

10 marks

Determine the exact destruction order. `class Employee { public: Employee() {} ~Employee() { cout << "Employee "; } }; class Manager { Employee e1; Employee e2; public: ~Manager() { cout << "Manager "; } };`

Your Answer

The destruction order is Manager->Employee->Employee because the derived object is destroyed first, followed by its member objects.

 Save Answer

Next →

Q7 Modification

10 marks

Write the minimum modification required to make Question safe for copying. class A { int* p; public: A(int x) : p(new int(x)) {} ~A() { delete p; } }; int main() { A a(10); A b(a); }

Your Answer

We need to add a copy constructor with deep copying so that each object gets its own dynamically allocated memory.

```
A(const A& other){  
    p=new int (*other.p);  
}
```

Save Answer

Next →

Q8 Find

10 marks

What happens if a class has a user-defined destructor but no explicitly declared move constructor?

Your Answer

If a class has a user defined destructor,the move constructor is not implicitly generated.Therefore move operations will not be automatically available

 Save Answer

Next →

Q9 **Replace**

10 marks

What should replace delete data in Question 55, assuming data points to an array?

Your Answer

Replace delete data with
delete[] data;
because data points to an array allocated using new[]

 Save Answer

Next →

Q10 Constructor

10 marks

Which constructor is called? `class Test { public: Test() { cout << "Default"; } Test(int x) { cout << "Parameterized"; } Test(const Test&) { cout << "Copy"; } }; int main() { Test a; Test b(10); Test c = b; }`

Your Answer

Test a calls the default constructor
Test b(10) calls the parameterized constructor
Test c=b calls the copy constructor

 Save Answer